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Feb 26, 2025

Overview

In this Git repo, I develop 6 lectures for the YSE Data Science Certificate team. Below, I outline the lecture material.

In these lectures, students begin analyzing data in R and then Python. A PDF of the lecture material accompanies each lecture so students can review it before. An incomplete notebook is provided so for them to synchronously write it during recorded lectures. A complete notebook is provided for reference. All examples are environmentally themed, as well as the problem sets.

These 6 lectures should included in the Foundations section of the certificate. I have a GitHub lecture may fit better in another section of the certificate.

Outline of Six Lectures

1. Thinking Like a Computer (45 – 60 min)

- Slides
 - a. what's programming
 - b. keyword definitions
 - c. problem definition and solution
 - d. why you should code (vs Excel and other software)
 - e. the importance of pseudo-code
 - f. a note on debugging

2. Base R, Part 1 (51 Minutes)

- Complete Jupyter Notebook
- Comments and Basic Data Types:
 - How to write comments in R using the # symbol.
 - The importance of comments for code readability and collaboration.
 - Basic data types in R: numeric (integer and double), character (string), and logical (Boolean).
- Basic Operations and Variables:
 - Performing basic arithmetic operations.
 - Creating and using variables.
 - Changing the value of variables.
- Vectors:
 - Creating and using vectors.
 - Understanding that vectors can only contain one data type.
- Matrices:
 - Creating matrices from vectors.
 - Understanding that matrices can only contain one data type.
- Lists:
 - Introduction to lists and their flexibility in storing different data types.
- Data Frames:
 - Creating data frames and understanding their structure.
 - Referencing columns, rows, and individual cells in a data frame.
 - Adding new columns to a data frame.
 - Assigning meaningful column names to a data frame.

3. Base R, Part 2

- Complete Jupyter Notebook

- Functions and Flow Control:
 - How to define and use functions in R.
 - Understanding and implementing loops (for loops).
 - Using if-else statements for conditional operations.
- Global vs Local Variables:
 - The difference between global and local variables.
 - How to use and access these variables within and outside functions.
- Error Handling:
 - Basic error handling techniques using `try()` and `tryCatch()` functions.
 - Strategies for debugging and handling errors effectively.
- Simulation:
 - Creating and running simulations using loops.
 - Example of simulating temperature changes over years.
- Apply Functions:
 - Using `apply()` function to perform operations on data frames.
 - Example of calculating mean temperatures for different parks.
- If-Else Statements:
 - Using if-else statements to correct data based on conditions.
 - Example of adjusting park distances based on gender.

4. Data Cleaning

- Complete Jupyter Notebook
- Introduction to Tidyverse:
 - Understanding what packages are and how to install and load them.
 - Overview of the Tidyverse collection of packages for data science.
- Data Manipulation with dplyr:
 - Using `mutate()` to create or modify columns.
 - Using `if_else()` for conditional operations.
 - Using `filter()` to subset data based on conditions.
 - Using `select()` to choose specific columns.
 - Using `group_by()` to group data and perform operations by a category.
 - Using `summarise()` to create summary tables.
- Practical Examples:
 - Examples of manipulating and cleaning data using the functions mentioned above.
 - Real-world scenarios such as adjusting park distances, filtering ecosystems by pollution levels, and summarizing species richness.

5. Introduction to Datasets

- Complete Jupyter Notebook
- Reading Data from CSV Files:
 - Using functions like `read_csv()` to load data from CSV files into R.
 - Understanding how to inspect the structure and summary of the dataset using functions like `glimpse()`, `summary()`, and `head()`.

6. Introduction to Data Visualization

- Complete Jupyter Notebook
- Revisits Data Cleaning and Preparation:
 - Revisits a few techniques for cleaning and preparing the dataset for analysis.
- Creating Plots with ggplot2:
 - Using the `ggplot2` package to create scatter plots and other types of visualizations.
 - Understanding the basic structure of a `ggplot` call, including specifying aesthetics with `aes()` and adding layers like `geom_point()`.
- Facet Wrapping:

- Learning how to use `facet_wrap()` to create multiple plots within the same graph, separated by a categorical variable (e.g., manufacturer).
- This allows for easy comparison of the same relationship across different categories.
- Customizing Plots:
 - Adding titles and labels to the plots using the `labs()` function to make the visualizations more informative.
 - Customizing other aspects of the plot to improve readability and presentation.
- Storing and Displaying Graphs:
 - Storing a graph as an object in R and displaying it when needed.
 - This helps in organizing and reusing plots efficiently.

7. Introduction to Python

- Complete Jupyter Notebook
- Understanding the similarities and differences between Python and other programming languages.
- Running basic Python code, such as printing “Hello World”.
- Basic Data Types and Operations:
 - Performing basic arithmetic operations.
 - Variable assignment and reassignment.
 - Understanding and using different data types: integers, floating-point numbers, strings, and Booleans.
 - Dynamic typing in Python.
- Data Structures:
 - Creating and manipulating lists, including lists with mixed data types.
 - Using NumPy arrays for numerical data and understanding their similarities to R matrices.
 - Creating and accessing elements in lists and arrays.
- DataFrames with Pandas:
 - Installing and importing the pandas package.
 - Creating DataFrames from NumPy arrays and dictionaries.
 - Renaming columns and accessing columns using both bracket `[]` notation and dot `.` notation.
 - Adding new columns to DataFrames.
- Indexing and Selecting Data:
 - Using `.iloc[]` for integer-location based indexing.
 - Using `.loc[]` for label-based indexing.
 - Accessing specific rows, columns, and cells using both `.iloc[]` and `.loc[]`.
- Functions:
 - Defining functions using the `def` keyword.
 - Writing a function to model a quadratic relationship and returning values.
- Loops:
 - Writing for loops to iterate over ranges and lists.
 - Combining loops with functions to perform repeated calculations.
- Conditional Statements:
 - Using if-else statements to create classifications based on conditions.
 - Iterating over DataFrame rows and using `.loc` to update DataFrame cells based on conditions.

8. Datasets and Python

- Complete Jupyter Notebook
- Data Manipulation with Pandas:
 - Creating and manipulating DataFrames.
 - Selecting specific columns using `loc` or `iloc`.
 - Grouping data and calculating aggregate statistics using `groupby()`.
 - Applying multiple aggregation functions using `agg()`.
 - Adding or modifying columns using `assign()`.
 - Conditionally modifying values using `np.where()`.
 - Filtering rows using boolean indexing.

- Dropping rows with missing values using `dropna()`.
- Reading in a Dataset and Gathering Basic Information:
 - Reading a CSV file into a DataFrame.
 - Displaying basic information about the DataFrame using methods like `head()`, `tail()`, `shape`, `columns`, `info()`, and `describe()`.
- Basic Data Cleaning:
 - Filtering and cleaning data based on specific research questions.
 - Dropping unnecessary columns.
- Basic Data Visualization:
 - Creating various plots using `matplotlib` and `seaborn`.

9. Data Cleaning, Part II

- Complete Jupyter Notebook
- `left_join()`
- `pivot_longer()`
- `pivot_wider()`

Problem Sets

1. Problem Set One
2. Problem Set Two
3. Problem Set Three